ATOCA Exercises

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Lemma 0.1 (ATOCA 3.38)**.** Let R be a ring and I, J be ideals of R. Assume J is finitely generated modulo I and assume that $J \subseteq \sqrt{I}$. Then, there exists some $n \geq 1$ such that $J^n \subseteq I$.

Proof. If J is contained I, this is trivial. So assume J is not contained in I.

Let $\bar{x}_1, \ldots, \bar{x}_n$ be the generators of J in R/I.

Since for all $i \in [n]$ we have that $x_i \in J$, for all $i \in [n]$ there exists some $k_i \geq 1$ such that $x_i^{k_i} \in I$.

Let $k = k_1 + \cdots + k_n$. By the binomial formula, J^k is contained in $\langle x_1^k, x_2^k, \ldots, x_n^k \rangle$.

But $\langle x_1^k, x_2^k, \ldots, x_n^k \rangle$ is the zero ideal in R/I by construction, so J^k is also the zero ideal, and we have the desired result. $\square$